Clay-CD-not-certified: Zero-Sorry Lean 4 Audit Certificate Demonstrating Why Linear Enstrophy-History Growth Does Not Certify BKM Blowup

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Abstract

Standalone zero-sorry Lean 4 negative audit certificate (‘lean/NotBlowup.lean‘) proving ‘bounded_M_keeps_bkm_finite‘ and ‘history_can_grow_while_bkm_stays_finite‘: a bounded vorticity supremum M keeps the Beale-Kato-Majda integral int_0^T M dt = M T finite on every finite interval [0, T] even while the quadratic enstrophy history integral int_0^T M^2 dt = M^2 T grows linearly. Consequently, linear growth of the history clock does not certify finite-time blowup or resolve Fefferman Statements (C) or (D).

1 Scope, Scientific Honesty, and Non-Claim Boundary

Negative audit certificate proving why a diverging quadratic history clock on [0, infty) is not a finite-time BKM blowup proof; explicitly declines to certify Fefferman (C) or (D).

- **Repository:** `https://github.com/BenFrohman/Clay-CD-not-certified`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23653015`
- **Earliest Git Commit SHA:** `bf14a32b0944c88da889ce99ceaee2901df53802` (2026-10-07 07:27:11 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Author: Benjamin Stanley Frohman
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This repository is not a resolution of Fefferman (C) or (D).

A request to title a repository as a Clay C/D resolution, and to certify a zero-sorry blowup proof from the Section 10 argument, is declined. The Section 10 materials do not prove that the Beale–Kato–Majda integral diverges in finite time.

Blowup in the BKM sense is divergence of ‖ ‖_∞ dt. The 2D and steady-Euler kill switch has M bounded, so _0^T M dt = M T stays finite, while M² grows. That is the opposite of a blowup proof.

Forced (C) and (D) are a different statement. An external claim exists in openai/NavierStokesAndEuler. This repository does not copy it and does not certify it.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/Clay-CD-not-certified` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	333 B	`c9c93bc0bf700801...`
`README.md`	819 B	`229b00a58e1bd3ef...`
`lean/NotBlowup.lean`	619 B	`a1e32fd19f20d0b3...`

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